

Two Fourier Peaks and a Moving Lattice Maximum

A proof of the OEIS A039831 asymptotic conjecture,
four correction orders, a smoothing hierarchy, and inverse growth

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October 1, 2026

Abstract

Let M_n be the largest coefficient of $P_n(q) = \prod_{j=1}^n (1 + q + q^3 + \cdots + q^{2j-1})$. The OEIS entry A039831 records Václav Kotešovec's 2023 conjecture $M_n \sim 3(n/e)^n$. We prove this statement and obtain explicit corrections through relative order n^{-4} , together with an algorithmic expansion to arbitrary algebraic order. Two features distinguish the answer from a routine Gaussian estimate. Integer rounding produces a nonconstant log-periodic term at order n^{-3} . A second Fourier neighborhood, at $q = -1$, produces a parity term at order n^{-4} , despite the exact equality of the total even- and odd-degree coefficient masses. This smaller contribution changes the maximizing exponent infinitely often; the number of failures of a single-neighborhood rounding rule up to N has order $\log N$. We prove a zero-multiplicity law for suppression of the second neighborhood under fixed Bernoulli smoothing, and derive Lambert- W inverse-growth formulas with the integer-threshold qualification made explicit. Exact coefficient computations through $n = 400$ and symbolic coefficient checks accompany the article. The arguments are self-contained mathematical proofs, not Lean-checked declarations. The source audit establishes the conjecture's current OEIS wording, not an exhaustive claim of publication priority.

Editorial note (October 1, 2026, batch 73O2). This report is manuscript 62 of batch 73 of ProveIt's incoming-reports intake, placed in the research-report collection in commit `6e193dd4f`. It is AI-assisted, unrefereed and not formalized. The text is the delivered manuscript; the write added the label prefix `tfp:`, notes marked “[Added October 1, 2026, batch 73O2]”, one bibliography entry, and Appendix C. The distinct-coefficient conjecture of A039824 is *not* resolved here.

Contents

1	Scope, provenance, and the main result	3
1.1	The problem and why the higher orders matter	3
1.2	Relationship to ProveIt and limits of the novelty claim	3
1.3	Notation and a first theorem	3
2	Independent summands and exact cumulants	4
2.1	Probability normalization	4
2.2	An exact cumulant algorithm	5
3	Why two Fourier neighborhoods are necessary	5
3.1	The neighborhood of $q = -1$	5
3.2	Fourier bounds with no hidden aperiodicity hypothesis	6
4	An arbitrary-order two-neighborhood expansion	7
4.1	Finite coefficient operators	7
4.2	What “all orders” means here	8
5	Derivation of the explicit coefficients	8
5.1	The contribution from zero	8
5.2	The contribution from π	9
6	The maximizing exponent and infinitely many rounding failures	9
6.1	Global localization of the mode	9
6.2	The parity-dependent transition	10
7	The log-periodic term and a limitation of power–log expansions	12
8	A fixed-smoothing hierarchy	12
9	Inverse growth and the integer threshold	13
9.1	A logarithmic core	13
9.2	Thresholds and a rigorous rounding qualification	14
10	Exact computation and reproducibility	15
10.1	A quadratic-work convolution step	15
10.2	Accuracy of the expansion	15
10.3	Inverse-index errors	16
11	Further research questions and topics	16
12	Proof status and suggested integration into ProveIt	17
12.1	What has been established	17
12.2	A reusable formalization order	17
A	A compact coefficient-generation specification	18
B	Source audit and package contents	18
C	Provenance and editorial notes	18

1 Scope, provenance, and the main result

1.1 The problem and why the higher orders matter

Write

$$P_n(q) = \prod_{j=1}^n \left(1 + \sum_{a=1}^j q^{2a-1} \right) = \sum_{k=0}^{n^2} c_{n,k} q^k, \quad M_n = \max_{k \in \mathbb{Z}} c_{n,k}, \quad (1.1)$$

with coefficients outside $[0, n^2]$ taken to be zero. This is the definition of OEIS A039831. Its record, inspected on October 1, 2026, still labels $M_n \sim 3n^n/e^n$ a conjecture and attributes it to Václav Kotešovec, January 5, 2023 [1]. The related entry A039824 concerns the *number of distinct coefficient values*, a different question; its conjectured eventual formula $n^2 - 3$ is not proved here [2].

The product in (1.1) is a simple, nonnegative, growing-support convolution model. Its total mass is factorial, its width is of order $n^{3/2}$, and its center has a harmonic correction. It is useful as an explicit test case for a general issue: a coefficient maximum is sensitive both to the integer lattice and to small Fourier contributions that a central limit theorem does not resolve. The leading conjecture alone is an application of local-limit methods. The substantive refinements below are the explicit lattice-resolved expansion, its mode transitions, and the suppression law under smoothing.

1.2 Relationship to ProveIt and limits of the novelty claim

The inspected ProveIt materials distinguish research arguments, executable checks, and the formal trusted theorem boundary [4]. This report follows that distinction. It connects to the repository’s transseries and inversion program, particularly its separation of a logarithmic core, remainder transport, and staircase inversion [5]. A targeted repository search for A039831 returned no match. This is not a claim that every file in the repository was examined.

The Fourier method itself is classical. Modern work of Dolgopyat and Hafouta explicitly treats trigonometric corrections and the instability of modular uniformity after deletion of finitely many summands [3]. Their stated setting uses uniformly bounded summands; our summands have growing support. We do not invoke their theorem outside its hypotheses. Instead we prove the required bounds directly and use that work to situate the mechanism. Targeted A-number searches did not identify an earlier proof of the particular conjecture or the refinements proved here. These searches are not a comprehensive literature review, and no global priority or “major breakthrough” claim is required for the mathematical statements.

1.3 Notation and a first theorem

Let

$$h_n = H_{n+1} = \sum_{j=1}^{n+1} \frac{1}{j}, \quad h_{2,n} = H_{n+1}^{(2)} = \sum_{j=1}^{n+1} \frac{1}{j^2}, \quad \mu_n = \frac{n(n-1)}{2} + h_n - 1, \quad \mathcal{E}_n = 3(n/e)^n. \quad (1.2)$$

The symbols $O(\cdot)$ below refer to $n \rightarrow \infty$; constants can depend on a fixed expansion order or a fixed bounded range of $k - \mu_n$. Define

$$a_1 = -\frac{431}{300}, \quad a_2 = \frac{23085971}{1764000}, \quad (1.3)$$

$$C_3(d; h, h_2) = -\frac{9}{2}d^2 - \frac{27}{2}h + \frac{9}{2}h_2 - \frac{107751433393}{1587600000}, \quad (1.4)$$

$$C_4(d; h, h_2) = \frac{1263}{40}d^2 - \frac{81}{2}d + \frac{3789}{40}h - \frac{1263}{40}h_2 + \frac{155775441709528423}{322727328000000}. \quad (1.5)$$

Theorem 1.1 (Four correction orders and the conjecture). *For every fixed $D > 0$, uniformly for integers k satisfying $|d| = |k - \mu_n| \leq D$,*

$$\begin{aligned} \frac{c_{n,k}}{\mathcal{E}_n} = & 1 + \frac{a_1}{n} + \frac{a_2}{n^2} + \frac{C_3(d; h_n, h_{2,n})}{n^3} \\ & + \frac{C_4(d; h_n, h_{2,n}) - 18(-1)^{n+k}}{n^4} + O_D\left(\frac{1 + \log^2 n}{n^5}\right). \end{aligned} \quad (1.6)$$

For all sufficiently large n , every maximizing exponent belongs to $\{\lfloor \mu_n \rfloor, \lceil \mu_n \rceil\}$. Consequently

$$M_n = 3(n/e)^n \left(1 - \frac{431}{300n} + \frac{23085971}{1764000n^2} + O\left(\frac{\log n}{n^3}\right) \right). \quad (1.7)$$

In particular, Kotešovec's stated asymptotic conjecture is true. The maximum of the two expressions in (1.6), evaluated at the indicated candidate exponents, approximates $M_n/\mathcal{E}_n$ with the same stated error, including at a mode transition.

The proof occupies Sections 2–6. Exact harmonic numbers in (1.6) are intentional: replacing them by their limits without expanding their tails changes later coefficients. The remainder is an asymptotic bound, not a numerical interval certificate with an explicit threshold.

2 Independent summands and exact cumulants

2.1 Probability normalization

Let X_j be independent, each uniform on the $j + 1$ integers

$$\{0, 1, 3, \dots, 2j - 1\}, \quad S_n = \sum_{j=1}^n X_j. \quad (2.1)$$

Evaluation of (1.1) at $q = 1$ gives

$$P_n(1) = (n + 1)!, \quad c_{n,k} = (n + 1)! \mathbb{P}(S_n = k). \quad (2.2)$$

The moment generating function of a single summand can be written

$$F_j(t) = 1 + \sum_{a=1}^j e^{(2a-1)t} = 1 + e^{jt} \frac{\sinh(jt)}{\sinh t}, \quad \mathbb{E}e^{tX_j} = \frac{F_j(t)}{j + 1}, \quad (2.3)$$

where the quotient is interpreted by continuity at zero.

Let $\kappa_{r,n}$ denote the r th cumulant of S_n , and put $V_n = \kappa_{2,n}$. Direct summation yields

$$\kappa_{1,n} = \mu_n, \quad (2.4)$$

$$V_n = \frac{n^3}{9} + \frac{n^2}{2} - \frac{29n}{18} + 3h_n - h_{2,n} - 2, \quad (2.5)$$

$$\kappa_{3,n} = n^2 - 6n + 14h_n - 9h_{2,n} + 2H_{n+1}^{(3)} - 7. \quad (2.6)$$

In particular $V_n \sim n^3/9$. This, together with factorial total mass, already explains the proposed factor 3.

2.2 An exact cumulant algorithm

For $r \geq 1$, set

$$m_r(j) = \frac{1}{j+1} \sum_{a=1}^j (2a-1)^r, \quad m_0(j) = 1. \quad (2.7)$$

The single-summand cumulants are generated by

$$b_r(j) = m_r(j) - \sum_{\ell=1}^{r-1} \binom{r-1}{\ell-1} b_\ell(j) m_{r-\ell}(j), \quad \kappa_{r,n} = \sum_{j=1}^n b_r(j). \quad (2.8)$$

This follows by differentiating $M'(t) = K'(t)M(t)$ for $M = e^K$ at zero. For example,

$$b_2(j) = \frac{j^2}{3} + \frac{2j}{3} - 2 + \frac{3}{j+1} - \frac{1}{(j+1)^2}, \quad (2.9)$$

$$b_3(j) = 2j - 7 + \frac{14}{j+1} - \frac{9}{(j+1)^2} + \frac{2}{(j+1)^3}, \quad (2.10)$$

$$b_4(j) = -\frac{2j^4}{15} - \frac{8j^3}{15} - \frac{4j^2}{5} + \frac{172j}{15} - 40 + \frac{93}{j+1} - \frac{83}{(j+1)^2} + \frac{36}{(j+1)^3} - \frac{6}{(j+1)^4}. \quad (2.11)$$

Thus all fixed-order cumulants are explicitly expressible using polynomials in n and generalized harmonic numbers.

Lemma 2.1 (Cumulant size and odd-order cancellation). *For each fixed $r \geq 2$, $\kappa_{r,n} = O_r(n^{r+1})$. For odd $r \geq 3$, the stronger estimate $\kappa_{r,n} = O_r(n^{r-1})$ holds. The analogous estimates hold for the second-neighborhood cumulants defined in Section 3.*

Proof. The first estimate follows either from (2.8) and $|X_j| \leq 2j$, or by applying Cauchy estimates to the logarithm of (2.3) on a disk of radius a sufficiently small constant divided by j . For the cancellation, set $t = z/j$ and $E(z) = z \coth z$, an even analytic function. Uniformly for small z ,

$$\frac{F_j(z/j)}{j+1} = \frac{e^z \sinh z/z}{1+1/j} \left(1 + \frac{E(z) - z}{j} - \frac{z^2}{6j^2} + O(j^{-4}) \right). \quad (2.12)$$

The zeroth-order logarithm has only the linear odd term z . The j^{-1} logarithmic term has only the linear odd term $-z/j$. All odd Taylor coefficients of degree at least three therefore first occur at order j^{-2} . Taking r derivatives with respect to t shows $b_r(j) = O_r(j^{r-2})$, and summation gives the assertion. For the signed factors used below, replace $E(z) - z$ by its negative and $1 + 1/j$ by $1 - 1/j$; the same cancellation applies. The extra fixed factor $(e^t - 1)/t$ has bounded cumulants and causes no difficulty. $\square$

3 Why two Fourier neighborhoods are necessary

3.1 The neighborhood of $q = -1$

Define an analytic function with value one at the origin by

$$B_n(t) = \frac{e^t - 1}{t} \prod_{j=2}^n \frac{\sum_{a=1}^j e^{(2a-1)t} - 1}{j-1}. \quad (3.1)$$

For $n \geq 2$, an exact factorization is

$$\frac{P_n(-e^t)}{(n+1)!} = \frac{(-1)^n t}{n(n+1)} B_n(t). \quad (3.2)$$

The factor t comes from $1 + q$. It is essential, not an error term. Write

$$\log B_n(t) = \nu_n t + \frac{U_n t^2}{2} + \sum_{r \geq 3} \lambda_{r,n} \frac{t^r}{r!}. \quad (3.3)$$

The function B_n is an analytic normalization, not a probability moment generating function. In particular some individual signed variances are negative. Nevertheless its aggregate quadratic coefficient is positive for all sufficiently large n , and direct calculation gives

$$\nu_n = \frac{n(n+3)}{2} - \frac{3}{2} + H_{n-1}, \quad (3.4)$$

$$U_n = \frac{n^3}{9} - \frac{n^2}{6} - \frac{41n}{18} + \frac{29}{12} - 3H_{n-1} - H_{n-1}^{(2)}, \quad (3.5)$$

$$\Delta_n := \nu_n - \mu_n = 2n - \frac{1}{2} - \frac{1}{n} - \frac{1}{n+1}. \quad (3.6)$$

The large displacement $\Delta_n \sim 2n$, though small relative to $\sqrt{V_n}$, converts the odd Fourier factor into a nonzero local contribution near the maximum.

For reproducibility, the signed moments for $j \geq 2$ are $\tilde{m}_0(j) = 1$ and $\tilde{m}_r(j) = \sum_{a=1}^j (2a-1)^r / (j-1)$ for $r \geq 1$. Use (2.8), then add the cumulants of $(e^t - 1)/t$, the moment generating function of the continuous uniform law on $[0, 1]$. This generates all the $\lambda_{r,n}$.

3.2 Fourier bounds with no hidden aperiodicity hypothesis

Put $\Phi_n(\theta) = P_n(e^{i\theta})/(n+1)!$. Fourier inversion gives

$$\mathbb{P}(S_n = k) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \Phi_n(\theta) e^{-ik\theta} d\theta. \quad (3.7)$$

The two ends near $\pm\pi$ are one neighborhood on the circle.

Lemma 3.1 (Two-neighborhood localization). *There are constants $a, c, C > 0$ such that, for all large n ,*

$$|\Phi_n(\theta)| \leq e^{-cn^3\theta^2} \quad (|\theta| \leq a/n), \quad (3.8)$$

$$|B_n(i\theta)| \leq Ce^{-cn^3\theta^2} \quad (|\theta| \leq a/n), \quad (3.9)$$

$$|\Phi_n(\theta)| \leq Ce^{-cn} \quad (\text{dist}(\theta, \pi\mathbb{Z}) \geq a/n). \quad (3.10)$$

The same bounds, with different constants, permit any fixed number of Taylor or Fourier-differentiation operations used below.

Proof. A single characteristic factor satisfies

$$\frac{F_j(i\theta)}{j+1} = \frac{1 + e^{ij\theta} \sin(j\theta)/\sin\theta}{j+1}. \quad (3.11)$$

For $|\theta| \leq a/n$, the identity $|\mathbb{E} e^{i\theta X_j}|^2 = \mathbb{E} \cos(\theta(X_j - X'_j))$, with X'_j an independent copy, and the inequality $1 - \cos u \geq c_0 u^2$ for small u show that the product is at most $\exp(-c_1 \theta^2 \sum_{j \geq j_0} j^2)$. This proves (3.8).

For the second neighborhood, the signed single-factor variance is

$$\tilde{b}_2(j) = \frac{j^2}{3} - \frac{2j}{3} - 2 - \frac{3}{j-1} - \frac{1}{(j-1)^2}. \quad (3.12)$$

It is at least a positive constant times j^2 for all sufficiently large j . Uniform analyticity on $|t| \leq a/j$ gives

$$\Re \log \left(\frac{\sum_{a=1}^j e^{(2a-1)i\theta}}{j-1} \right) = -\frac{1}{2} \tilde{b}_2(j) \theta^2 + O(j^4 \theta^4).$$

Choose a sufficiently small. The error is then absorbed into the negative quadratic term. Finitely many small factors and $(e^{i\theta} - 1)/(i\theta)$ contribute only bounded factors, proving (3.9).

For completeness, set $\rho = \text{dist}(\theta, \pi\mathbb{Z})$. If $\rho \geq K/n$ for a sufficiently large fixed K , then, for $j \geq n/2$,

$$\left| \frac{F_j(i\theta)}{j+1} \right| \leq \frac{1 + 1/|\sin \theta|}{j+1} \leq \frac{2}{n} + \frac{\pi}{K} < q < 1.$$

If instead $a/n \leq \rho \leq K/n$, put $u = n\rho$ and $v = j/n$. On the compact rectangle $u \in [a, K]$, $v \in [1/2, 1]$, $|\sin(uv)/(uv)|$ has supremum strictly less than one. Formula (3.11), with its $O(1/j)$ extra term, gives the same strict bound for all sufficiently large n . At least a positive proportion of the factors thus have modulus at most $q < 1$. The remaining factors are characteristic functions and have modulus at most one. This proves (3.10). Fixed derivatives merely add polynomial factors, which the exponential bounds absorb away from the centers; inside the centers they are handled by the displayed analytic estimates. $\square$

Remark 3.2 (Exact parity balance does not remove the second peak). Since $P_n(-1) = 0$, the sums of the even- and odd-degree coefficients are exactly equal. But (3.2) has a simple zero, not an identically vanishing neighborhood. A local coefficient expansion sees its derivatives. Removing the first Bernoulli factor destroys the exact parity balance and leaves a Fourier value of magnitude $2/[n(n+1)]$. This is a concrete instance of the stability issue discussed in [3].

4 An arbitrary-order two-neighborhood expansion

4.1 Finite coefficient operators

Let

$$G_V(x) = \frac{1}{\sqrt{2\pi V}} e^{-x^2/(2V)}, \quad \text{He}_r(x) = (-1)^r e^{x^2/2} \frac{d^r}{dx^r} e^{-x^2/2}. \quad (4.1)$$

For a nonnegative integer R , use multi-indices $\mathbf{m} = (m_3, \dots, m_{R+2})$ with

$$w(\mathbf{m}) = \sum_{r=3}^{R+2} (r-2)m_r \leq R, \quad D(\mathbf{m}) = \sum_{r=3}^{R+2} r m_r. \quad (4.2)$$

An empty index set when $R = 0$ means the single zero multi-index. For a cumulant list $\boldsymbol{\eta}$, define

$$T_R(x; \boldsymbol{\eta}, V) = \sum_{w(\mathbf{m}) \leq R} \left[\prod_{r=3}^{R+2} \frac{\eta_r^{m_r}}{m_r! (r!)^{m_r}} \right] \frac{\text{He}_{D(\mathbf{m})}(x/\sqrt{V})}{V^{D(\mathbf{m})/2}}, \quad (4.3)$$

$$S_R(x; \boldsymbol{\eta}, V) = \sum_{w(\mathbf{m}) \leq R} \left[\prod_{r=3}^{R+2} \frac{\eta_r^{m_r}}{m_r! (r!)^{m_r}} \right] \frac{\text{He}_{D(\mathbf{m})+1}(x/\sqrt{V})}{V^{(D(\mathbf{m})+1)/2}}. \quad (4.4)$$

These are finite, directly computable expressions.

Theorem 4.1 (All algebraic orders). *For every fixed nonnegative integer R , uniformly for all integer k ,*

$$\begin{aligned} \mathbb{P}(S_n = k) &= G_{V_n}(k - \mu_n) T_R(k - \mu_n; \boldsymbol{\kappa}_n, V_n) \\ &\quad + \frac{(-1)^{n+k}}{n(n+1)} G_{U_n}(k - \nu_n) S_R(k - \nu_n; \boldsymbol{\lambda}_n, U_n) \\ &\quad + O_R(V_n^{-1/2} n^{-(R+1)/2}). \end{aligned} \quad (4.5)$$

The assertion is an absolute-error estimate, not a relative estimate at arbitrary tail indices. Once the two-candidate mode localization is established, it gives the maximum to arbitrary relative algebraic order by increasing R and taking the maximum of the two candidate values.

Proof. Split (3.7) into the neighborhoods from Lemma 3.1. The complement is exponentially small. At zero, center the characteristic function and put $u = \theta\sqrt{V_n}$. Its logarithm is

$$-\frac{u^2}{2} + \sum_{r=3}^{R+2} \frac{\kappa_{r,n}}{r!V_n^{r/2}}(iu)^r + \text{remainder}. \quad (4.6)$$

The coefficient for index r is $O_r(n^{-(r-2)/2})$. Expanding the exponential by this weight gives exactly the multi-indices in (4.3).

Here are the remainder details. Fix R , and first restrict $|u| \leq \sqrt{K \log n}$, where K is a sufficiently large fixed number. Uniform analytic cumulant bounds on $|t| \leq a/n$ bound the logarithmic Taylor remainder by $C_R n^{-(R+1)/2}(1 + |u|^{R+3})$ on that interval for all large n . The omitted terms of the weighted exponential expansion have the same order times a fixed polynomial in $|u|$, after multiplication by e^{-cu^2} . Integration removes that polynomial without introducing a power of $\log n$. Outside the cutoff, the Gaussian bounds make the integral smaller than the required power by taking K larger. This proves an integrated error $O_R(n^{-(R+1)/2})$, before the Jacobian $V_n^{-1/2}$.

Each retained integral is evaluated using

$$\frac{1}{2\pi} \int_{\mathbb{R}} (i\theta)^r e^{-V\theta^2/2 - ix\theta} d\theta = G_V(x) \frac{\text{He}_r(x/\sqrt{V})}{V^{r/2}}. \quad (4.7)$$

Extending the integrals to the real line changes them only by the already controlled Gaussian tails. All error estimates are uniform in x because $|e^{-ix\theta}| = 1$.

At the other neighborhood, use (3.2). Fourier inversion contributes $(-1)^k$, and the exact factor supplies $(-1)^n/[n(n+1)]$ and one additional $i\theta$. The same argument with U_n and $\lambda_{r,n}$ therefore gives (4.4). Its own remainder is $O_R(n^{-2}U_n^{-1}n^{-(R+1)/2})$, smaller than the displayed total remainder. This proves (4.5). $\square$

4.2 What “all orders” means here

The theorem is not a claim of convergence of an infinite asymptotic series. For every desired algebraic error, it supplies a finite formula and a proved remainder. Using (2.8), polynomial division, Faulhaber’s formulas, and Euler–Maclaurin expansion of harmonic numbers turns the finite formula into powers of n^{-1} with logarithmic coefficients and bounded lattice variables. Constants can involve γ and values $\zeta(r)$.

For bounded $k - \mu_n$, half-integer powers in the standardized Hermite notation combine into integer powers: odd Hermite polynomials supply an additional odd factor of their small argument. The second neighborhood has the same property, with $k - \nu_n = -2n + O(1)$. Taking a maximum preserves an error bound but need not preserve a single smooth coefficient function across a mode transition. We establish neither exponentially improved sectors nor Borel summability. Those stronger meanings of “full transseries” are left open.

5 Derivation of the explicit coefficients

5.1 The contribution from zero

Use Lemma 2.1 in (4.5). Up to relative order n^{-4} at bounded $d = k - \mu_n$, the only odd-cumulant term needed is $-\kappa_{3,n}d/(2V_n^2)$. Even terms are generated by the partitions of weights at most four using $\kappa_4, \kappa_6, \kappa_8, \kappa_{10}$, where κ_{2r} has weight $r - 1$. As examples, the central factors through weight three are

$$1 + \frac{\kappa_4}{8V^2} - \frac{\kappa_6}{48V^3} + \frac{35\kappa_4^2}{384V^4} + \frac{\kappa_8}{384V^4} - \frac{7\kappa_4\kappa_6}{128V^5} + \frac{385\kappa_4^3}{3072V^6}. \quad (5.1)$$

Here and in the following calculation the subscript n is suppressed. For nonzero d , use the full Hermite polynomial, not merely its constant term.

The relevant polynomial portions of the cumulants begin as follows:

$$\kappa_4 = -\frac{2n^5}{75} - \frac{n^4}{5} - \frac{26n^3}{45} + \frac{26n^2}{5} + O(n), \quad (5.2)$$

$$\kappa_6 = \frac{16n^7}{441} + \frac{8n^6}{21} + \frac{104n^5}{63} + O(n^4), \quad (5.3)$$

$$\kappa_8 = -\frac{16n^9}{135} - \frac{8n^8}{5} + O(n^7), \quad (5.4)$$

$$\kappa_{10} = \frac{256n^{11}}{363} + O(n^{10}). \quad (5.5)$$

These formulas follow from (2.8); the accompanying symbolic script performs the exact rational operations.

The factorial and variance factors are

$$\frac{(n+1)!}{\sqrt{2\pi} n^{n+3/2} e^{-n}} = (1+n^{-1}) \exp\left(\frac{1}{12n} - \frac{1}{360n^3} + O(n^{-5})\right), \quad (5.6)$$

$$\frac{n^{3/2}}{3\sqrt{V}} = \left(1 + \frac{9}{2n} - \frac{29}{2n^2} + \frac{27h_n - 9h_{2,n} - 18}{n^3}\right)^{-1/2}. \quad (5.7)$$

Multiplying the Hermite contributions, $e^{-d^2/(2V)}$, and these two factors gives $1 + a_1/n + a_2/n^2 + C_3/n^3 + C_4/n^4$. For a quick check on the first coefficient,

$$a_1 = \underbrace{\frac{13}{12}}_{\text{factorial}} - \underbrace{\frac{9}{4}}_{\text{variance}} - \underbrace{\frac{27}{100}}_{\text{fourth cumulant}} = -\frac{431}{300}. \quad (5.8)$$

The first lattice loss is $-d^2/(2V) \sim -9d^2/(2n^3)$, and the first skew displacement term is $-\kappa_3 d/(2V^2) \sim -81d/(2n^4)$. These are useful independent checks on (1.4) and (1.5).

5.2 The contribution from π

For bounded d and $x = k - \nu_n = d - \Delta_n$, the leading term from the second neighborhood is

$$\frac{(-1)^{n+k}}{n(n+1)} G_{U_n}(x) \frac{x}{U_n}. \quad (5.9)$$

Now $x = -2n + O(1)$, $U_n \sim n^3/9$, and $G_{U_n}(x)/G_{V_n}(0) = 1 + O(n^{-1})$. Therefore

$$\frac{\text{second-neighborhood contribution}}{G_{V_n}(0)} = -\frac{18(-1)^{n+k}}{n^4} + O(n^{-5}). \quad (5.10)$$

The quartic correction changes this only at the next order, and the odd-cumulant bounds in Lemma 2.1 control the remaining terms. The normalization from (5.6) and (5.7) is $1 + O(n^{-1})$, so the same coefficient -18 appears in (1.6). Taylor expansion of the remaining explicit factors, keeping h_n and $h_{2,n}$ exact, gives a uniform remainder $O_D((1 + \log^2 n)n^{-5})$. This proves the local expansion part of Theorem 1.1.

6 The maximizing exponent and infinitely many rounding failures

6.1 Global localization of the mode

A local expansion at a proposed maximum does not by itself prove that this is the global maximum. We supply that step separately.

Lemma 6.1 (Principal smooth peak and secondary bound). *Let $A_n(d)$ be the real-valued contribution of $|\theta| \leq a/n$ to (3.7), centered so that $d = k - \mu_n$. It is defined for real d . For a sufficiently small fixed $\varepsilon > 0$, it has a unique critical point d_n^* in $|d| \leq \varepsilon\sqrt{V_n}$, and*

$$d_n^* = -\frac{\kappa_{3,n}}{2V_n} + O(n^{-2}) = -\frac{9}{2n} + O(n^{-2}). \quad (6.1)$$

On that interval A_n is strictly concave, with curvature comparable to $-G_{V_n}(0)/V_n$. If $B_n^{\text{arc}}(d)$ denotes the other neighborhood's contribution, including its parity factor at integer indices, then

$$|B_n^{\text{arc}}(d)| \leq CG_{V_n}(0) \frac{|d| + n}{n^5} \quad (|d| \leq \varepsilon\sqrt{V_n}). \quad (6.2)$$

Proof. Differentiate the principal Fourier integral. The proof of Theorem 4.1 applies with additional powers of θ . Uniformly on the stated interval, its second derivative is

$$A_n''(d) = -\frac{G_{V_n}(0)}{V_n}(1 + O(\varepsilon^2 + n^{-1})).$$

At zero, even cumulants contribute nothing to the first derivative. The cubic term and Lemma 2.1 give

$$A_n'(0) = G_{V_n}(0) \left(-\frac{\kappa_{3,n}}{2V_n^2} + O(n^{-5}) \right).$$

Strict concavity and the intermediate value theorem produce the unique critical point and (6.1).

For the other arc, the extra $i\theta$ differentiates its smooth Gaussian profile. Its even-cumulant terms are odd in $x = d - \Delta_n$ and bounded, on a fixed standardized interval, by $CG_{V_n}(0)|x|/U_n$. The odd-cumulant terms contribute at most $CG_{V_n}(0)n^{-4}$ by Lemma 2.1. Multiplication by $1/[n(n+1)]$, use of $|x| \leq |d| + Cn$, and sufficiently many terms of (4.5) give (6.2). $\square$

Proposition 6.2 (Two integer candidates). *For all sufficiently large n , every global maximizing exponent lies in $\{\lfloor \mu_n \rfloor, \lceil \mu_n \rceil\}$.*

Proof. The uniform absolute-error expansion in Theorem 4.1 first implies a uniform local central limit estimate with error $o(G_{V_n}(0))$. Comparing with an integer at distance at most $1/2$ from μ_n confines every maximizer to $|d| \leq \varepsilon\sqrt{V_n}$, for any fixed small ε and all large n .

Within this interval, compare to the integer nearest μ_n . Any integer outside the two stated candidates has $|d| \geq 1$, while the comparison integer has distance at most $1/2$. Lemma 6.1, with ε chosen small enough that the upper and lower curvature bounds have ratio arbitrarily close to one, shows a principal-arc loss at least

$$cG_{V_n}(0) \frac{1 + d^2}{n^3} \quad \text{for such a noncandidate.}$$

The displacement $d_n^* = O(n^{-1})$ does not alter this fixed quadratic gap. The total secondary-arc advantage is at most $CG_{V_n}(0)(|d| + n)n^{-5}$, which is uniformly smaller. The minor-arc error is exponentially small. Thus a noncandidate cannot maximize. $\square$

Proposition 6.2 completes the proof of Theorem 1.1: both candidate offsets are bounded, and taking a maximum is Lipschitz in the sup norm of the two errors.

6.2 The parity-dependent transition

Put

$$m = \lfloor \mu_n \rfloor, \quad r_n = \mu_n - m - \frac{1}{2}, \quad s_n = (-1)^{n+m}. \quad (6.3)$$

Subtract (1.6) at m and $m + 1$. The harmonic and constant terms cancel. Since $2(m - \mu_n) + 1 = -2r_n$,

$$\frac{c_{n,m+1} - c_{n,m}}{\mathcal{E}_n} = \frac{9r_n}{n^3} + \frac{-\frac{1263}{20}r_n - \frac{81}{2} + 36s_n}{n^4} + O\left(\frac{1 + \log^2 n}{n^5}\right). \quad (6.4)$$

In the transition window $r_n = O(n^{-1})$, this becomes

$$\frac{c_{n,m+1} - c_{n,m}}{\mathcal{E}_n} = \frac{9}{n^3} \left(r_n - \frac{\frac{9}{2} - 4s_n}{n} + O\left(\frac{1 + \log^2 n}{n^2}\right) \right). \quad (6.5)$$

Thus the transition is at $r_n \approx 1/(2n)$ when $s_n = 1$, but at $r_n \approx 17/(2n)$ when $s_n = -1$. Rounding the principal smooth mode alone would instead put it at $9/(2n)$ for both parities. Formula (6.5) decides the sign whenever the separation from the proposed boundary dominates its displayed error. It is not an exact finite- n rounding rule inside the unresolved boundary layer.

Theorem 6.3 (The smaller Fourier contribution changes the mode infinitely often). *Let $\hat{k}_n$ be a nearest integer to $\mu_n - 9/(2n)$, with any fixed tie convention. Then $\hat{k}_n$ fails to maximize $c_{n,k}$ for infinitely many n . The same assertion holds for rounding the exact principal smooth peak $\mu_n + d_n^*$. If $E(N)$ counts failures of the explicit rule $\hat{k}_n$ up to N , then, for all sufficiently large N ,*

$$\log N - O(1) \leq E(N) \leq 12 \log N + O(1). \quad (6.6)$$

In particular the failures have order $\log N$ and natural density zero.

Proof. Let $b_n = H_{n+1} - 1$ and, for each sufficiently large integer j , let n_j be the first index with $b_{n_j} > j + 1/2$. The overshoot is between zero and $1/(n_j + 1)$ because $b_n - b_{n-1} = 1/(n + 1)$. For $n = n_j + \ell$, $\ell = 1, 2, 3$, therefore

$$\frac{1 - o(1)}{n} \leq r_n \leq \frac{4 + o(1)}{n}, \quad \lfloor b_n \rfloor = j. \quad (6.7)$$

Also

$$s_n = (-1)^{n + \lfloor \mu_n \rfloor} = (-1)^{n(n+1)/2 + j}.$$

The parity of $n(n + 1)/2$ has period four, with two occurrences of each parity in each period. In any three consecutive indices there is an $s_n = 1$. At such an index in (6.7), the true transition boundary $1/(2n)$ lies below r_n by a fixed multiple of $1/n$, whereas the principal boundary $9/(2n)$ lies above it by a fixed multiple of $1/n$. The true maximizing index is $m + 1$ and the principal rounding rule chooses m . All margins exceed the error in (6.5); the $O(n^{-2})$ correction in (6.1) also cannot change this conclusion.

There are $\log N + O(1)$ crossings up to N , since $b_N = \log N + \gamma - 1 + o(1)$, and the three-index windows are eventually disjoint. This proves the lower bound, allowing an $O(1)$ loss for the last window.

For the upper bound, (6.4) and (6.5) imply that a failure is possible only when $0 < r_n < 10/n$, for all sufficiently large n ; outside this interval both rules choose the same candidate. Such an index lies just after a half-integer crossing. If it is $n = n_j + \ell$, then $r_n \geq \ell/(n + 1)$, so $r_n < 10/n$ permits at most twelve indices per crossing. Finitely many earlier indices are absorbed into $O(1)$. $\square$

Remark 6.4. Rounding the mean itself also fails infinitely often. In the first three indices at or immediately after a half-integer crossing, select $s_n = -1$. The overshoot is at most $(3 + o(1))/n$, below the true $17/(2n)$ boundary, even though the nearest integer to the mean is already the upper one. No equidistribution assumption about harmonic numbers or exponential fractional parts is used in either argument.

7 The log-periodic term and a limitation of power–log expansions

Set $\delta(x) = \text{dist}(x, \mathbb{Z})$ and

$$K_0 = -\frac{27}{2}\gamma + \frac{9}{2}\zeta(2) - \frac{107751433393}{1587600000}. \quad (7.1)$$

The nearest-integer loss at third order is independent of which of two almost tied candidates eventually wins. Indeed, a departure from the nearest-mean candidate occurs only within $O(1/n)$ of a tie and changes the squared distance by $O(1/n)$. Using $h_n = \log n + \gamma + O(1/n)$ and $h_{2,n} = \zeta(2) + O(1/n)$ in Theorem 1.1 gives

Corollary 7.1 (Explicit log-periodic correction).

$$\frac{M_n}{\mathcal{E}_n} = 1 + \frac{a_1}{n} + \frac{a_2}{n^2} + \frac{-\frac{27}{2}\log n + K_0 - \frac{9}{2}\delta(\log n + \gamma - 1)^2}{n^3} + O\left(\frac{\log n}{n^4}\right). \quad (7.2)$$

There is no polynomial Q for which the same expression can be replaced by $1 + a_1/n + a_2/n^2 + Q(\log n)/n^3 + o(n^{-3})$.

Proof. The distance-to-integers function, and its square on the bounded range $[0, 1/2]$, are Lipschitz. Replacing the exact harmonic argument by $\log n + \gamma - 1$ therefore changes the third-order term by $O(n^{-4})$. This proves (7.2).

For the second assertion, a possible polynomial would have to equal $-(27/2)x$ plus a constant and an $o(1)$ error after the bounded remainder is isolated. But for $a = 0$ and $a = 1/2$, integers n_j nearest $\exp(j + 1 - \gamma + a)$ satisfy $\log n_j + \gamma - 1 - j \rightarrow a$. The squared distance has the two distinct limits 0 and $1/4$. A single constant cannot absorb both. $\square$

The periodic term has the absolutely convergent Fourier representation

$$\delta(x)^2 = \frac{1}{12} + \frac{1}{\pi^2} \sum_{\ell=1}^{\infty} \frac{(-1)^\ell}{\ell^2} \cos(2\pi\ell x). \quad (7.3)$$

It follows by integrating twice to compute the Fourier coefficients of x^2 on $[-1/2, 1/2]$. Thus the third-order correction can alternatively be represented using complex powers $n^{2\pi i \ell}$. It is periodic in $\log n$ with period one, or multiplicatively periodic at leading order under $n \mapsto en$. This oscillation is generated by taking a lattice maximum, not by a singularity of the original probability model. The additional parity term at fourth order has a different origin, namely the second Fourier neighborhood.

8 A fixed-smoothing hierarchy

The role of the simple zero at -1 can be varied without changing the growing part of the model. For a fixed integer $r \geq 0$, define

$$P_{n,r}(q) = (1+q)^r \prod_{j=2}^n \left(1 + \sum_{a=1}^j q^{2a-1}\right). \quad (8.1)$$

Its total mass is $2^{r-1}(n+1)!$. Let $\mu_{n,r}$ and $V_{n,r}$ be the mean and variance after normalization. Then

$$\mu_{n,r} = \mu_n + \frac{r-1}{2}, \quad V_{n,r} = V_n + \frac{r-1}{4}. \quad (8.2)$$

The original sequence is $r = 1$.

Theorem 8.1 (Zero multiplicity and the second-neighborhood order). *For fixed $r \geq 0$ and bounded $k - \mu_{n,r}$, the contribution of the π neighborhood to the normalized coefficient probability, divided by $G_{V_{n,r}}(0)$, is*

$$(-1)^{n-1+r+k} \begin{cases} 2^{1-r} 3^r (-1)^{r/2} (r-1)!! n^{-2-3r/2} (1 + O(n^{-1})), & r \text{ even}, \\ 2^{2-r} 3^{r+1} (-1)^{(r+1)/2} r!! n^{-(3r+5)/2} (1 + O(n^{-1})), & r \text{ odd}. \end{cases} \quad (8.3)$$

The convention is $(-1)!! = 1$. In particular the first orders are n^{-2} for $r = 0$, n^{-4} for $r = 1$, and n^{-5} for $r = 2$.

Proof. The exact analogue of (3.2) is

$$\frac{P_{n,r}(-e^t)}{2^{r-1}(n+1)!} = \frac{(-1)^{n-1+r} 2^{1-r} t^r}{n(n+1)} B_{n,r}(t), \quad (8.4)$$

where

$$B_{n,r}(t) = \left(\frac{e^t - 1}{t} \right)^r \prod_{j=2}^n \frac{\sum_{a=1}^j e^{(2a-1)t} - 1}{j-1}.$$

Its center is $\nu_n + (r-1)/2$, its quadratic coefficient is $U_n + (r-1)/12$, and the displacement from the principal center is still Δ_n . The t^r factor gives the r th Hermite derivative of the Gaussian. At $x = -2n + O(1)$ the standardized argument tends to zero at rate $n^{-1/2}$.

For even r , use $\text{He}_r(0) = (-1)^{r/2} (r-1)!!$ and $U^{-r/2} \sim 3^r n^{-3r/2}$. For odd r , use $\text{He}_r(z) = (-1)^{(r-1)/2} r!! z + O(z^3)$, with $z = x/\sqrt{U} \sim -6n^{-1/2}$. Combine with $2^{1-r}/[n(n+1)]$. The Gaussian ratio is $1 + O(n^{-1})$. Even-cumulant corrections have relative order n^{-1} or smaller; the odd-cumulant cancellation from Lemma 2.1 makes those corrections smaller still. This proves (8.3). $\square$

For $r = 0$ the secondary contribution is larger than the ordinary n^{-3} rounding loss. Accordingly the two-candidate mode theorem for $r = 1$ must not be transferred to $r = 0$ without a new argument. For $r \geq 2$ the parity effect occurs later. The theorem isolates a general principle: a finite smoothing factor can move a Fourier contribution across several orders, and odd zero multiplicity has an additional cancellation near the moving center.

9 Inverse growth and the integer threshold

9.1 A logarithmic core

From (1.7), taking logarithms gives

$$\log(M_n/3) = n(\log n - 1) + \frac{A}{n} + \frac{B}{n^2} + O\left(\frac{\log n}{n^3}\right), \quad A = -\frac{431}{300}, \quad B = \frac{5907087}{490000}. \quad (9.1)$$

The second constant is $B = a_2 - a_1^2/2$; confusing a multiplicative coefficient with a logarithmic coefficient would give the wrong inverse.

For large y , let

$$L = \log(y/3), \quad x = \frac{L}{W_0(L/e)}. \quad (9.2)$$

The principal real Lambert function is characterized by $W_0(z)e^{W_0(z)} = z$. It follows directly that $x(\log x - 1) = L$. This is the same core-and-correction organization used in the repository's inversion materials [5]. [Added October 1, 2026, batch 73O2: in the canonical volume *Transseries and Inversion* [6] the exact solution of this core is its theorem “Exact solution of the dominant block” (`p0:thm:lambert-core`), and the two corrections of Theorem 9.1 are an instance of its

theorem “Perturbed inversion” (`p0:thm:perturbed-inversion`). The manuscript re-derives them for its own forward map, which is a pure power carrier $(n/e)^n$ rather than a gamma function. No novelty is claimed for the inversion method; what is specific to this report is the forward input A , B and the lattice-dependent third coefficient (9.4).]

Theorem 9.1 (Inverse expansion on the sequence values). *When $y = M_n$ and x is given by (9.2),*

$$n = x + \frac{431}{300x \log x} - \frac{5907087}{490000x^2 \log x} + O(x^{-3}). \quad (9.3)$$

The error concerns the recovered real index at the exact sequence values. It is not an $o(1)$ approximation to a staircase at arbitrary arguments without rounding.

Proof. Let $F(t) = t(\log t - 1)$, so $F'(t) = \log t$. Equation (9.1) first gives $n - x = O(1/(x \log x))$ by the mean value theorem. Write $n = x + \varepsilon$ and expand:

$$0 = (\log x)\varepsilon + \frac{A}{x} + \frac{B}{x^2} + O\left(\frac{\log x}{x^3}\right).$$

Here the quadratic Taylor term and the changes in A/n and B/n^2 are smaller than the stated remainder. Division by $\log x$ gives (9.3). $\square$

The next logarithmic coefficient, retaining a chosen candidate offset d and exact harmonic quantities, is

$$-\frac{9}{2}d^2 - \frac{27}{2}h_n + \frac{9}{2}h_{2,n} - \frac{197099473}{3937500}. \quad (9.4)$$

Thus further inversion is possible, but the lattice dependence must be retained or treated by candidate envelopes. It cannot be replaced by an invented smooth third coefficient.

9.2 Thresholds and a rigorous rounding qualification

The coefficients are nonnegative, and the next factor has constant coefficient one, so $M_{n+1} \geq M_n$. Formula (1.7) also gives $M_{n+1}/M_n \sim n$, hence strict increase eventually. Define

$$N(y) = \min\{n \geq 1 : M_n \geq y\}. \quad (9.5)$$

Let $t(y)$ be the large real solution of

$$F(t) + A/t + B/t^2 = \log(y/3). \quad (9.6)$$

The left side is strictly increasing for large t . There is a constant $C > 0$ such that, for all sufficiently large y ,

$$\lceil t(y) - Ct(y)^{-3} \rceil \leq N(y) \leq \lceil t(y) + Ct(y)^{-3} \rceil. \quad (9.7)$$

To prove this, transport the $O(\log n/n^3)$ error in (9.1) through a derivative bounded below by a positive multiple of $\log n$. An index smaller than $t - Ct^{-3}$ cannot reach y , while an integer at least $t + Ct^{-3}$ must reach it. Consequently $N(y) = \lceil t(y) \rceil$ away from the indicated small neighborhoods of integers. No exact ceiling rule is asserted in the boundary layer. [Added October 1, 2026, batch 73O2: this bracket is an instance of the volume’s staircase theorem (`p0:thm:staircase` [6]), whose part (1) gives $N(y) = \lceil \nu(y) \rceil$ for every admissible increasing interpolation ν and whose part (2), the separation condition, decides the ceiling from an approximation x with $|x - \nu(y)| \leq \eta$ exactly when $\lceil x - \eta \rceil = \lceil x + \eta \rceil$. The separation step and its failure for arbitrarily small η are formalized in Lean as `Fabius.staircase_separation` and `Fabius.staircase_separation_fails` in `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`. Those declarations are general; they do not verify any statement of this report.]

For a smooth approximation $\widetilde{M}_J(t)$ with relative error $O(t^{-J})$, the same argument gives an index error $O(t^{-J}/\log t)$ on matched values, provided its logarithmic derivative is comparable to $\log t$. For the all-order result here, this principle is applied to the two lattice-resolved candidate expressions rather than to a nonexistent globally smooth power-log series for the maximum.

10 Exact computation and reproducibility

10.1 A quadratic-work convolution step

Let $a_k = c_{n-1,k}$, extended by zero. Multiplication by the n th factor requires

$$c_{n,k} = a_k + s_k, \quad s_k = \sum_{j=1}^n a_{k-(2j-1)}. \quad (10.1)$$

The sliding recurrence

$$s_k = s_{k-2} + a_{k-1} - a_{k-2n-1} \quad (10.2)$$

uses separate running sums for the two parities. It computes a whole row in $O(n^2)$ integer additions and subtractions, hence rows through N in $O(N^3)$ such operations, with $O(N^2)$ live integer entries. This is an arithmetic-operation count, not a bit-complexity bound: the integers grow with N .

The supplied program computes every row through $n = 400$ using arbitrary-precision integers, checks total mass against the exact integer $(n+1)!$, and matches the first twenty maximum values printed in A039831 [1]. The record's b-file was not needed for these calculations. An independent naive convolution on small rows and exact moment identities are included as unit tests. Symbolic checks derive the four coefficients independently from the cumulant recurrence and Hermite integration. These finite checks support implementation correctness; they do not replace the analytic proofs.

10.2 Accuracy of the expansion

The table reports (approximation $-M_n$)/ M_n . “Order j ” means (1.6) through n^{-j} at the exactly computed maximizing exponent, including the second neighborhood for order four.

n	Leading only	Order two	Order three	Order four
50	2.497×10^{-2}	8.835×10^{-4}	-1.218×10^{-4}	2.010×10^{-5}
100	1.336×10^{-2}	1.240×10^{-4}	-8.562×10^{-6}	7.063×10^{-7}
200	6.921×10^{-3}	1.701×10^{-5}	-6.047×10^{-7}	2.581×10^{-8}
400	3.525×10^{-3}	2.313×10^{-6}	-3.933×10^{-8}	8.016×10^{-10}

The computations use high-precision floating point only after exact integer coefficients have been obtained. The displayed errors are numerical measurements, not outward-rounded interval bounds.

The separate program `all_orders.py` evaluates the finite formula of Theorem 4.1 directly, with exact rational cumulants before numerical Gaussian–Hermite evaluation. At $n = 100$, $k = 4954$, and weight budget $R = 8$, its relative error is about 1.146×10^{-10} . Retaining exact cumulants and variance in this finite expression automatically retains some higher-order terms that are absent from the four-power truncation. Increasing the order at a fixed n is not guaranteed to improve an asymptotic approximation.

Examples in the exact calculation where rounding $\mu_n - 9/(2n)$ chooses the wrong maximizing exponent include $n = 139, 140, 141, 142$ and $n = 373, 374, 379, 380$. At $n = 139$, $\mu_n \approx 9595.522425$: principal rounding selects 9595, but the actual unique maximizing exponent is 9596. The first-order parity threshold itself is asymptotic: for instance its unqualified prediction at $n = 56$ is wrong. This does not contradict (6.5), which retains a boundary-layer error and asserts no universal small- n rule.

10.3 Inverse-index errors

At the exact input $y = M_n$, the errors in the Lambert core and in (9.3) are:

n	$x - n$	Two-correction inverse minus n
50	-6.3043×10^{-3}	-1.9121×10^{-4}
100	-2.8811×10^{-3}	-2.3115×10^{-5}
200	-1.3017×10^{-3}	-2.7890×10^{-6}
400	-5.8723×10^{-4}	-3.3910×10^{-7}

The complete integer maxima, modes, means, relative errors, and inverse checks are supplied in CSV files. Reproduction requires no network access once the package's Python dependencies are installed.

11 Further research questions and topics

The following problems are not claimed solved by this article.

1. Effective thresholds and interval certificates. Make the constants in the two-neighborhood bounds explicit, prove a numerical threshold for Proposition 6.2, and join it to exhaustive exact verification below that threshold. This would turn an eventual theorem into a fully certified mode algorithm for all n . The main difficulty is obtaining useful, not merely enormous, constants in the analytic remainder.

2. Exact mode-transition counting. Theorem 6.3 proves only upper and lower logarithmic bounds. Is there a constant c with $E(N) \sim c \log N$? The leading four-periodic picture suggests four failures near a generic half-integer harmonic crossing, but near-boundary overshoots require control before this can be a theorem. Determining whether those exceptional overshoots affect the leading counting constant is a natural quantitative problem.

3. The distinct-coefficient conjecture A039824. Resolve the conjectured formula $n^2 - 3$ for all $n > 6$ [2]. The present central analysis can constrain nearby equal coefficients but does not rule out distant coincidences across the whole row. A proof would need additional exact algebra or a global coefficient-ordering argument. This problem must not be conflated with asymptotic knowledge of the largest coefficient.

4. General fixed smoothing polynomials. Replace $(1 + q)^r$ in (8.1) by a fixed polynomial $Q(q)$ with nonnegative coefficients. Determine a complete expansion in terms of the order of vanishing and the initial Taylor coefficients of Q at roots of unity. Theorem 8.1 gives the simplest one-root instance, including its odd-multiplicity cancellation.

5. Growing smoothing. Let $r = r(n)$ grow. Determine the regimes in which suppression of the secondary neighborhood competes with the change in the principal variance and with the lattice spacing. The proof for fixed r is not uniform in r and cannot be reused without new estimates.

6. More residue classes and several resonant neighborhoods. For products supported on zero plus a growing arithmetic progression, classify which roots of unity contribute at algebraic orders. Develop a mode-selection rule that compares several residue classes rather than only two parities. A useful objective is a finite-state lattice-envelope calculus for asymptotic maxima.

7. Exponentially improved expansions. After the two algebraic neighborhoods have been expanded to high order, identify the next exponential scales, optimal truncation indices, and possible Stokes effects under complex deformation. The exponentially small minor-arc bound in this article is not an identification of those scales and is not a resurgent description.

8. Formal verification of the analytic-to-discrete bridge. Formalize the product normalization, cumulant recurrence, Fourier integral, and two-neighborhood estimates, then verify the coefficient arithmetic and mode comparison. The most reusable component would be a theorem transporting uniform local expansions into a finite candidate set for a discrete maximum. Such a theorem would apply beyond this particular OEIS entry.

9. Certified inverse thresholds with lattice data. Build an inverse algorithm that uses the Lambert core to find a small candidate set of indices and evaluates rigorously bounded two-sector formulas there. Prove exact decisions except in explicitly marked intervals, falling back to exact coefficient computation when needed. This would implement the distinction between a real inverse approximation and an integer staircase in Section 9. [Added October 1, 2026, batch 73O2: the candidate-set step is the volume’s proposition “Certified discrete inversion” (`p0:prop:certified-rounding` [6]); what remains open is the rigorous two-sector bound that feeds it.]

12 Proof status and suggested integration into ProveIt

12.1 What has been established

The paper proves the conjectured leading asymptotic for A039831, the uniform four-order local expansion, an arbitrary-order coefficient operator with a remainder, eventual two-candidate localization, parity-dependent transitions, infinitely many failures of principal rounding with logarithmic counting bounds, the nonexistence of a pure power-log third-order coefficient for the maximum, the fixed-smoothing law, and the displayed inverse-growth and threshold statements.

These are ordinary mathematical proofs in the article. They have not been checked by Lean, Rocq, or an independent human referee in this work. The exact computations and symbolic assertions are executable checks of finite claims, not a formalization of the asymptotic theorems. The phrases “proved here” and “current OEIS conjecture” have these separate meanings throughout.

12.2 A reusable formalization order

A suitable first layer would define the finite coefficient product and prove the sliding recurrence, total mass, and probability normalization. A second layer would formalize the raw-moment/cumulant recurrence and the exact formulas for $\mu_n, V_n, \kappa_{3,n}$. The symbolic identities in the supplied script are natural candidates for exact rational reflection certificates.

The analytic layer would prove the two-neighborhood localization and Hermite integral identity, followed by a fixed-order expansion theorem. The mode-localization argument should be a separate theorem with explicit hypotheses: uniform central-limit localization, a concave principal profile, and a secondary-profile bound. Finally the repository’s logarithmic-core and remainder-transport organization can be used for inversion, with its staircase caveat preserved [5]. [Added October 1, 2026, batch 73O2: that is, the volume’s `p0:thm:lambert-core`, `p0:thm:perturbed-inversion` and `p0:thm:staircase` [6]; see the notes in Section 9.] Existing repository module descriptions are a roadmap, not evidence that these new hypotheses have already been formalized.

A A compact coefficient-generation specification

The symbolic derivation in the package follows this finite recipe. Generate $b_r(j)$ by (2.8) for $r \leq 10$ and sum their polynomial parts. For even cumulants use nonnegative exponents (e_4, e_6, e_8, e_{10}) satisfying

$$e_4 + 2e_6 + 3e_8 + 4e_{10} \leq 4. \quad (\text{A.1})$$

Put $D = 4e_4 + 6e_6 + 8e_8 + 10e_{10}$ and add

$$\prod_{r \in \{4,6,8,10\}} \frac{\kappa_r^{e_r}}{e_r!(r!)^{e_r}} \frac{\text{He}_D(d/\sqrt{V})}{V^{D/2}}. \quad (\text{A.2})$$

For this precision, retaining the constant and quadratic terms of each even Hermite polynomial suffices. Add the single odd term $-\kappa_3 d/(2V^2)$. Multiply by $e^{-d^2/(2V)}$, the variance factor (5.7), and the factorial factor (5.6). Expand through n^{-4} , leaving $h_n, h_{2,n}$ as exact symbols. Finally add $-18(-1)^{n+k}/n^4$.

The omitted odd terms begin at relative order n^{-5} : $\kappa_5 d/V^3 = O(n^{-5})$ and $\kappa_3^2/V^3 = O(n^{-5})$ are the first examples. The proper rational parts of $b_4(j)$ sum to $O(\log n)$ and first affect this calculation at $O(\log n/n^6)$. These order checks explain why polynomial cumulant parts suffice for the four coefficients, although not for arbitrary order.

B Source audit and package contents

The source check was performed on October 1, 2026. The live A039831 internal record displayed revision #9, January 5, 2023, with the conjecture still present. A039824 displayed its separate conjectures. The ProveIt recursive-tree response inspected at the beginning of the work identified commit `a5ad4ac2e2d4e642a6c015836a70492d40fcbc71`. Relevant repository reads were the root README and the consolidated transseries/inversion README; targeted searches included the A039831 identifier and asymptotic inversion. Search indices and a live branch need not share a commit, so the mathematical proofs above do not depend on an unverified assertion about repository-wide coverage.

The accompanying archive contains the editable L^AT_EX source, compiled PDF, an exact coefficient and numerical verification program, a symbolic coefficient derivation, an arbitrary-order finite-formula evaluator, finite unit tests, numerical and inverse CSV outputs, a source manifest, and build instructions. No external paper or font file is redistributed. The bibliography supplies the primary sources used for context and provenance. [Added October 1, 2026, batch 7302: in the shipped report the scripts are in `code/`, their recorded outputs and the requirements file in `data/`, the compiled PDF is a new build of this text, and the delivered README is replaced by the report README; Appendix C lists what is shipped.]

C Provenance and editorial notes

[Added October 1, 2026, batch 7302.]

Source. The report was built from one manuscript, manuscript 62 of batch 73 of ProveIt's incoming-reports intake: the archive `A039831_Two_Fourier_Peaks.zip` (main file `article.tex`, an 18-page PDF). It arrived in commit `aa43cc555` and was placed in commit `6e193dd4f`, which deleted the archive from `docs/incoming`; it survives in `aa43cc555`. Its pin is commit `a5ad4ac2e` (the full hash is printed in the previous section), an ancestor of the placement commit; the two repository files the manuscript read, the root README and the transseries volume's README, are unchanged between the pin and the placement.

What the write changed.

- Every label gained the prefix `tfp::`; none was otherwise renamed or removed. The manuscript had 74 labels.
- Added: the editorial note after the abstract; bracketed notes in Section 9, research question 9, Section 12 and Appendix B; one bibliography entry [6]; and this section.
- Nothing was removed. Every non-claim of the manuscript is kept: A039824 is not proved; the remainders are asymptotic, not interval certificates; the numerical errors are measurements; the first-order mode prediction fails at $n = 56$; the Dolgopyat–Hafouta theorem is not invoked; no global priority, referee review or formal verification is claimed.

A039824 is not resolved. The report proves Kotešovec’s conjecture on the *largest* coefficient (A039831). The *number of distinct coefficient values* (A039824), conjectured to be $n^2 - 3$ for $n > 6$, is a different question, and nothing here settles it.

Checks by the intake. The intake recomputed the rows exactly through $n = 200$ ($M_1, \dots, M_5 = 1, 2, 4, 13, 57$, row masses $(n + 1)!$) and found the relative error of the two-correction formula (1.7) to be about 8.8×10^{-4} , 1.24×10^{-4} and 1.70×10^{-5} at $n = 50, 100, 200$, a ratio of about 7.3 per doubling, consistent with $O(\log n/n^3)$. It reran the three programs on a copy; their outputs equal the shipped ones apart from line endings and the elapsed-time line. These checks do not replace an independent review of the proofs.

References

- [1] The OEIS Foundation, *A039831: Largest coefficient in expansion of $\prod_{i=1}^n (1+q+q^3+\dots+q^{2i-1})$* . Entry by Olivier Gérard; asymptotic conjecture by Václav Kotešovec, January 5, 2023. Inspected October 1, 2026. <https://oeis.org/A039831>; internal record: <https://oeis.org/A039831/internal>.
- [2] The OEIS Foundation, *A039824: Number of different coefficient values in the same product*. Conjectured eventual formula by Ralf Stephan, March 7, 2004; related generating-function and recurrence conjectures by Colin Barker, May 2, 2017. Inspected October 1, 2026. <https://oeis.org/A039824>.
- [3] D. Dolgopyat and Y. Hafouta, *Edgeworth expansions for independent bounded integer valued random variables*, Stochastic Processes and their Applications **152** (2022), 486–532. <https://arxiv.org/abs/2011.14852>. The abstract and publication metadata were consulted for context; the theorem is not invoked for our growing-support summands.
- [4] V. Reshetnikov, *ProveIt*, root README and repository tree, inspected October 1, 2026. <https://github.com/VladimirReshetnikov/ProveIt>.
- [5] V. Reshetnikov, ProveIt research corpus, *Transseries: the polynomial–logarithmic calculus and its inversions*, consolidated volume README, inspected October 1, 2026. https://github.com/VladimirReshetnikov/ProveIt/blob/main/Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/README.md.
- [6] [Added in the batch-73O2 write.] V. Reshetnikov and contributors, *Transseries and Inversion*, the volume whose README is [5]: its source `transseries_and_inversion.tex` lies beside that README. Cited here: Part “Inversion: the apparatus for a rapidly growing function” (state of commit `6e193dd4f`).